Mike Verwer

Applied Data Professional

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M.Sc. mathematician transitioning into data analytics, pairing quantitative depth, technical communication, and a hands-on automation portfolio. Top-of-class Data Analytics diploma (4.33 GPA). Python, SQL, ML, Tableau, local GenAI tooling.

Experience

MATHEMATICS & PHYSICS TUTOR | C2 Education

Dec 2025 - May 2026 | Vancouver, BC.

- Tutored mathematics across grades 5–12 and taught grade 11 physics, adapting explanations across a wide range of skill levels.
- Delivered introductory Python instruction, building foundational coding and computational-thinking skills in a beginner student.

MATHEMATICS & STATISTICS INSTRUCTOR | Mohawk College, Engineering Dept.

2019-2020; 2022-2025 | Hamilton, ON.

- Delivered instruction in statistical analysis, probability theory, and quantitative methods to 150+ engineering and business students per semester.
- Designed and implemented data-driven assessment strategies to track student performance metrics, identifying learning trends and adjusting curriculum to improve pass rates by estimated 15%.
- Communicated complex mathematical and statistical concepts to diverse, non-technical audiences through data visualization and real-world case studies.

QUANTITATIVE FINANCE INSTRUCTOR | Humber College, Liberal Arts & Science

2022-2023 | Etobicoke, ON.

- Instructed 60+ students in financial mathematics, time-value analysis, and statistical modeling for business applications.

TECHNICAL AUTHOR | Cengage Learning – Calculus: Early Transcendentals, 9th Ed.

2019 | Remote

- Developed LaTeX-based mathematical solutions and documentation for international textbook publication demonstrating proficiency in technical typesetting, version control, and quality assurance processes.

Projects

JOB-SEARCH AGENT | n8n, Ollama, Docker, JavaScript, REST APIs

2026

- Built a fully local, end-to-end agentic workflow that runs daily: fetches new postings via the JSearch job API, parses and sanitizes them, scores each against defined criteria using a locally hosted LLM (Qwen2.5 7B via Ollama), and emails a ranked daily digest of top matches.
- Orchestrated in self-hosted n8n (Docker), integrating Ollama through host.docker.internal with custom Code nodes, JSON sanitization, and HTTP request handling.
- Designed a hybrid scoring architecture. Location and seniority get handled deterministically from a centralised config, while the LLM produces structured JSON output for skill match and growth signal assessment.
- Weighted final scores across five configurable dimensions with hard disqualifier gates. A deduplication layer filters already-seen postings across daily runs using n8n's persistent Static Data store.

Skills

PROGRAMMING

Proficient:
Python • SQL • LaTeX

Experienced:
JavaScript • CSS • HTML • C#

Familiar:
Java • C++

LIBRARIES/FRAMEWORKS

Pandas • Scikit-Learn • NumPy • Seaborn • Matplotlib • Plotly.js • BeautifulSoup • Tkinter • PySide6 • SymPy • OpenPyXL

TOOLS/PLATFORMS

Git • Docker • MS SQL Server • MySQL • MongoDB • Tableau • Power BI • Excel • n8n • Ollama

Education

DOUGLAS COLLEGE

PDD Data Analytics

May 2025 - 2026 | New Westminster, BC.

Computing Studies & Information Systems

4.33/4.33 GPA

MCMASTER UNIVERSITY

Master's in Mathematics

Sept 2016 - July 2019 | Hamilton, ON.

Faculty of Science

Universal Algebra, Formal Logic

MCMASTER UNIVERSITY

Honours Bachelor's in Mathematics & Statistics

Sept 2013 - May 2016 | Hamilton, ON.

Faculty of Science

Projects (Continued)

CONFIGURABLE ML PIPELINE FRAMEWORK & DATASET DIAGNOSIS | Python, Scikit-Learn, imbalanced-learn, Pandas

2026

- Built a reusable pipeline factory wrapping Scikit-Learn `ColumnTransformer` and `GridSearchCV`, running 4 classifiers across 7 preprocessing configurations (scaling, PCA, SMOTE, SelectKBest) from a single tagged-step interface.
- Kept SMOTE oversampling and scaling inside cross-validation folds to prevent data leakage, evaluating out-of-fold via StratifiedKFold.
- Prioritized class-1 recall over accuracy based on the cost asymmetry of false negatives in medical-risk prediction, using a custom F1 scorer for hyperparameter selection.
- Diagnosed the dataset as signal-free (no feature-target dependence, AUC near 0.5 across all models) rather than overfitting to noise, attempting feature engineering before concluding the data was likely synthetic.

CAMPUS AD-POSTING DATABASE | MS SQL Server, MySQL, T-SQL

2026

- Designed and implemented a relational database from scratch for a campus ad-posting platform, including system analysis, EER diagram creation, and schema mapping.
- Applied normalization through BCNF, documenting a deliberate exception for a transitive dependency (Employee's OfficeLocation → Extension) rather than fully decomposing it.
- Produced platform-specific DDL scripts for both MS SQL Server and MySQL, with standardized constraint naming conventions and seed data.
- Implemented business-rule validation directly in stored procedures for every entity lifecycle (role grants, ad review and posting workflow, board capacity, messaging), enforcing state-transition and referential rules beyond what foreign keys alone capture.
- Built a library of over a dozen views handling reporting and aggregation (board capacity, review queues, message counts), so that read access does not require hand-written joins against the base schema.

THE CYBER THREAT LANDSCAPE | Tableau, Python (Pandas)

2026

- Built an interactive Tableau dashboard on twelve years of publicly reported cyber incidents (15,400+ events after filtering) from the University of Maryland's CISSM/GoTech Cyber Events Database, with a Pandas script resolving category-label inconsistencies (casing, spacing, pluralization) in the raw feed before load.
- Used a RANK() table calculation to surface top-3 country labels on paired choropleth maps, log-scaling the color encoding to keep the map legible against a distribution where one country accounts for nearly half of all targeted events.
- Decomposed a multi-valued motive field into eight CONTAINS()-based indicator flags to avoid fragmenting compound categories, and built a parameter-driven Actor/Target map toggle using Tableau's dashboard zone-visibility feature to swap views without breaking mark interactivity.
- Scoped three dashboard filter actions individually (one excluding six dependent sheets) to prevent self-referential filtering, and applied source-level data filters, with documented rationale, to exclude a partial reporting year and unclassified records.

PAGEMAKER – AUTOMATED STATIC SITE GENERATOR | Python, BeautifulSoup, XML, Tkinter

2024 - 2026

- Engineered a Python-based static site generator to automate HTML page creation from templates, reducing manual development time by 80%.
- Currently powers my personal website mikeverwer.github.io.
- Implemented HTML parsing and modification using BeautifulSoup to dynamically inject content from markdown files.
- Generated XML sitemaps for SEO optimization and created a GUI interface for streamlined content management.

CENTRAL LIMIT THEOREM SIMULATOR | Python, PySimpleGUI, NumPy, Matplotlib

2024

- Built an interactive simulator using dice rolls with constraint-preserving controls (auto-normalizing sliders that sum to a fixed total under user-locked values) and custom canvas hit-detection, computing exact theoretical distributions via iterative NumPy convolution and validating them against a live Monte Carlo simulation with synchronized, click-to-inspect graphs.